Real Time Applications

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Introduction to Apache Spark:

- Apache Spark is an open-source distributed computing system designed for fast and general-purpose cluster computing.
- Developed at UC Berkeley's AMPLab and later donated to Apache Software Foundation.
- Provides an interface for programming entire clusters with implicit data parallelism and fault tolerance.
- Spark is known for its speed, achieving this primarily through in-memory data processing, minimizing disk I/O, and optimized query execution.

Spark Ecosystem

Spark provides multiple components for different data processing tasks:

Component	Description
Spark Core	The foundation of Spark, responsible for task scheduling, memory management, fault recovery, and distributed task execution.
Spark SQL	Module for structured data processing using SQL queries and DataFrames.
Spark Streaming	Enables real-time stream processing of live data streams.
Spark MLlib	Machine Learning library built on top of Spark for scalable algorithms.
Spark GraphX	Library for graph processing and computation.

Components of the Spark Unified Stack:

- **Driver Program** – Runs the main function of the application and defines distributed datasets on the cluster.
- **Cluster Manager** – Allocates resources to Spark applications.
- **Worker Nodes** – Perform actual computation using executors.
- **Executors** – Run the tasks assigned by the driver and return results.